

Milestone 2: AI-Assisted Threat Analysis Report

PCAP Threat Detection Project

1. Introduction

Project Overview

This report documents the AI-assisted network traffic investigation carried out for Milestone 2 of the PCAP Threat Detection Project. The objective was to analyze network traffic extracted from a PCAP capture using AI-assisted techniques, and to identify potentially suspicious communication patterns, unusual network relationships, protocol usage, and indicators that warrant further security investigation.

The analysis combined PCAP-derived CSV network traffic data, AI-assisted traffic investigation using ChatGPT, manual analyst review and validation, and security-focused interpretation of the AI-generated observations. The purpose of using AI was not to automatically classify traffic as malicious, but to help a security analyst identify important traffic patterns and generate investigation hypotheses for further review.

Source Dataset

Original capture: 2021-08-16-formbook.pcap

The PCAP data was converted into CSV format for AI-assisted analysis. Because of dataset size limitations and analysis efficiency requirements, the extracted CSV dataset was divided into three segments.

Dataset Segment	File Name	Records Analyzed
Segment 1	traffic_segment_1.csv	1,200 packets
Segment 2	traffic_segment_2.csv	1,200 packets
Segment 3	traffic_segment_3.csv	1,188 packets

Total analyzed traffic: 3,588 network records

Investigation Objectives and Analysis Principle

The AI-assisted investigation focused on identifying frequently communicating source and destination IP addresses, classifying internal/private and external/public IP addresses, detecting repeated communication patterns, and identifying possible automated behaviour or beaconing indicators. It also covered unusual external communication relationships, port and protocol usage, large data transfers and unusual frame sizes, and traffic requiring additional SOC investigation.

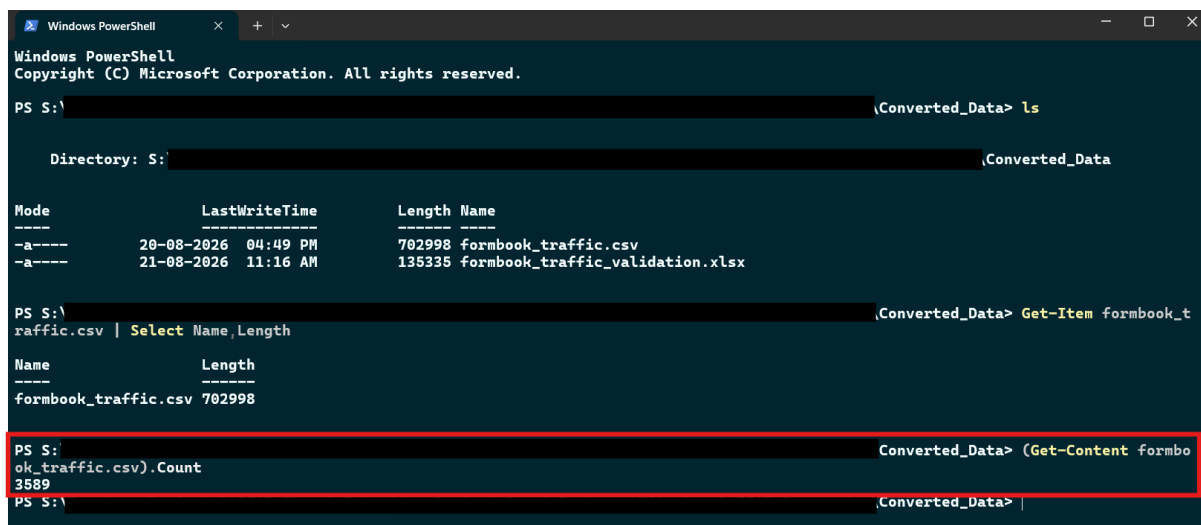
The AI analysis was treated as an investigation assistance tool rather than a decision-making tool. Traffic patterns identified by the AI were considered indicators requiring review, investigation hypotheses, and areas for further validation; the analysis did not automatically classify any communication as malicious.

Analysis approach: Observation → Evidence → Security Relevance → Validation Requirement

2. AI-Assisted Analysis Methodology

2.1 Data Preparation and Segmentation

The PCAP capture was converted into CSV format containing the network traffic information required for analysis. The extracted fields included source and destination IP addresses, protocol information, source and destination ports, packet and frame sizes, and communication relationships. The dataset was validated, then separated into three smaller datasets: traffic_segment_1.csv, traffic_segment_2.csv, and traffic_segment_3.csv. Segmentation improved AI processing reliability, allowed independent analysis of different traffic portions, made it possible to compare findings across multiple sections of the capture, and supported a combined analysis after each segment had been reviewed individually.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS S:\> (Converted_Data> ls

Directory: S:\> (Converted_Data> ls

Mode                LastWriteTime         Length Name
----                -
-a----          20-08-2026   04:49 PM          702998 formbook_traffic.csv
-a----          21-08-2026   11:16 AM         135335 formbook_traffic_validation.xlsx

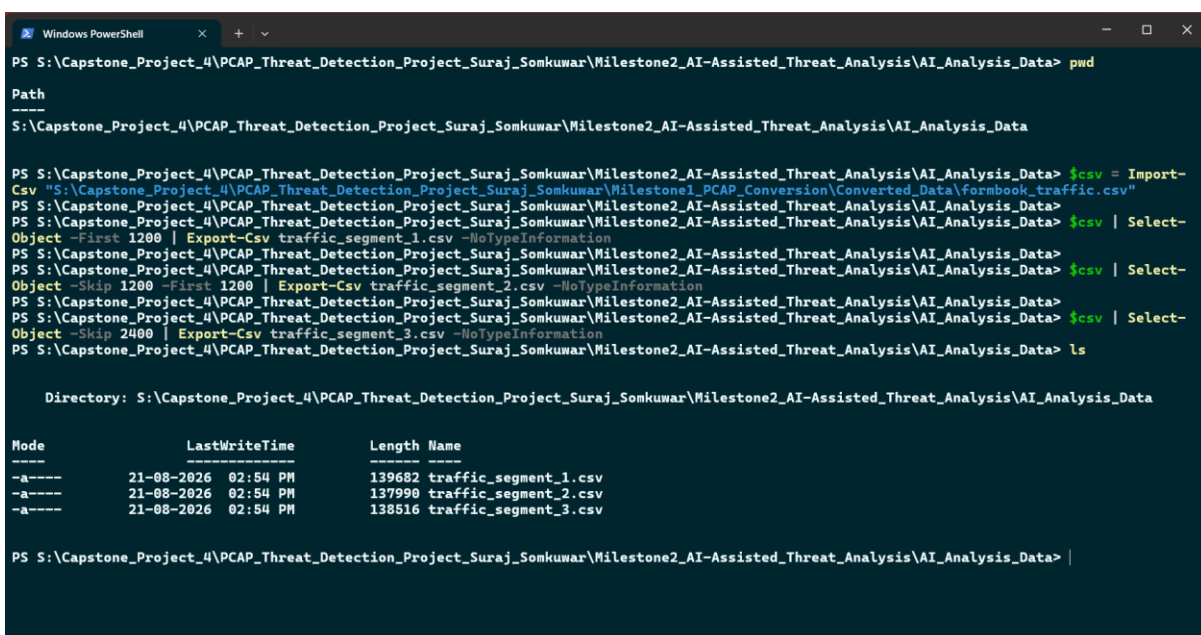
PS S:\> (Converted_Data> Get-Item formbook_t
raffic.csv | Select Name,Length

Name                Length
----                -
formbook_traffic.csv 702998

PS S:\> (Converted_Data> (Get-Content formbo
ok_traffic.csv).Count
3589
PS S:\> (Converted_Data> |
```

Figure 1: CSV Row Count Validation

Validation of the extracted CSV dataset before segmentation.



```
Windows PowerShell
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> pwd

Path
----
S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data

PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> $csv = Import-
Csv "S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone1_PCAP_Conversion\Converted_Data\formbook_traffic.csv"
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data>
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> $csv | Select-
Object -First 1200 | Export-Csv traffic_segment_1.csv -NoTypeInformation
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data>
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> $csv | Select-
Object -Skip 1200 -First 1200 | Export-Csv traffic_segment_2.csv -NoTypeInformation
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data>
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> $csv | Select-
Object -Skip 2400 | Export-Csv traffic_segment_3.csv -NoTypeInformation
PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> ls

Directory: S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data

Mode                LastWriteTime         Length Name
----                -
-a----          21-08-2026   02:54 PM         139682 traffic_segment_1.csv
-a----          21-08-2026   02:54 PM         137990 traffic_segment_2.csv
-a----          21-08-2026   02:54 PM         138516 traffic_segment_3.csv

PS S:\Capstone_Project_4\PCAP_Threat_Detection_Project_Suraj_Somkuwar\Milestone2_AI-Assisted_Threat_Analysis\AI_Analysis_Data> |
```

Figure 2: CSV Segment Creation

Creation of the three CSV segments used during AI-assisted analysis.

2.2 AI Analysis Process

Each CSV segment was submitted separately to ChatGPT with a cybersecurity-focused investigation prompt. The AI was instructed to analyze communication frequency, IP classification, repeated traffic behaviour, external communication relationships, protocol usage, large data transfers, and possible security concerns, and to separate confirmed observations from assumptions, avoid declaring traffic malicious without evidence, and recommend additional SOC validation steps. Figures 3 through 8 record the prompt submitted and the response generated for each of the three segments.

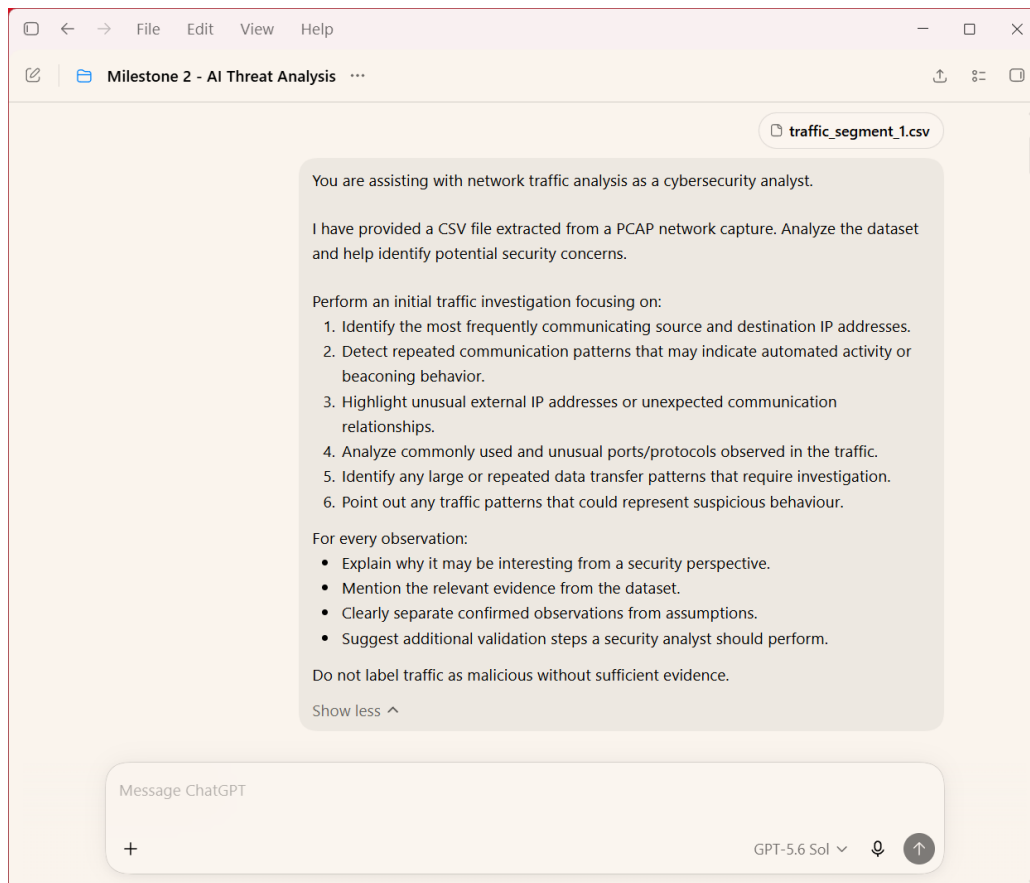


Figure 3: Segment 1 AI Investigation Prompt

Initial AI prompt used to analyze Segment 1 traffic.

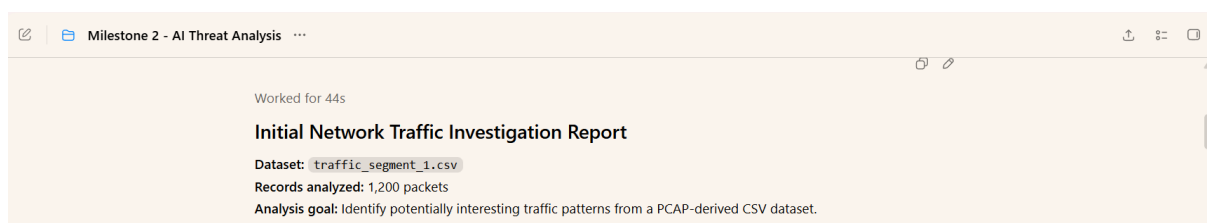


Figure 4: Segment 1 AI Generated Response

AI-generated initial traffic investigation response for Segment 1.

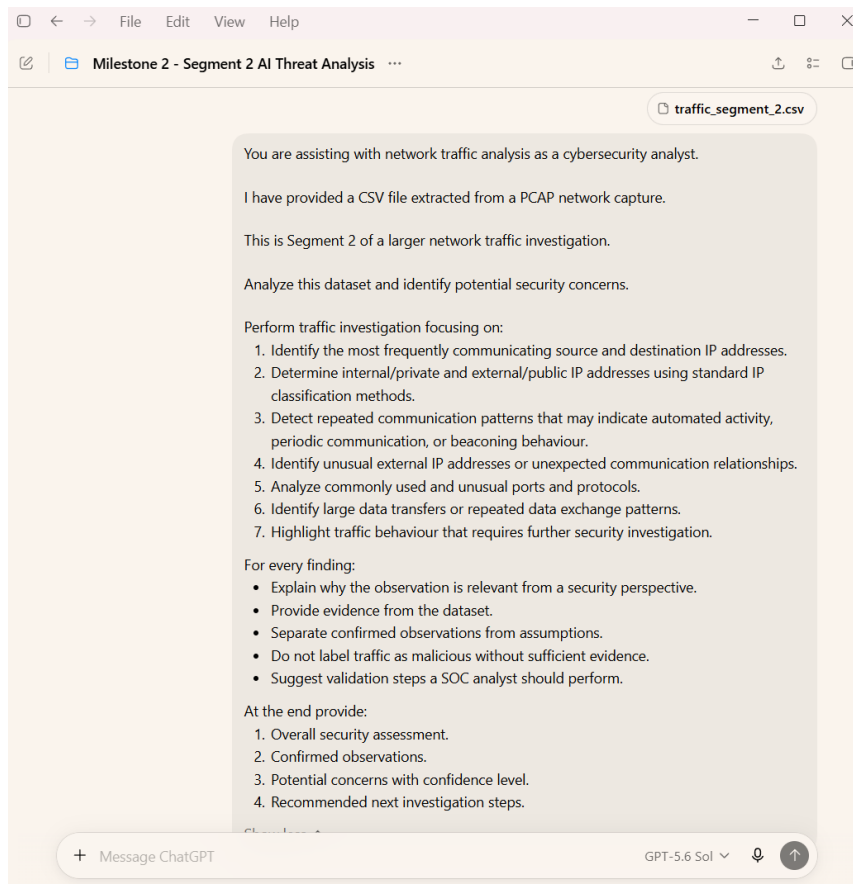


Figure 5: Segment 2 AI Investigation Prompt



Figure 6: Segment 2 AI Generated Response

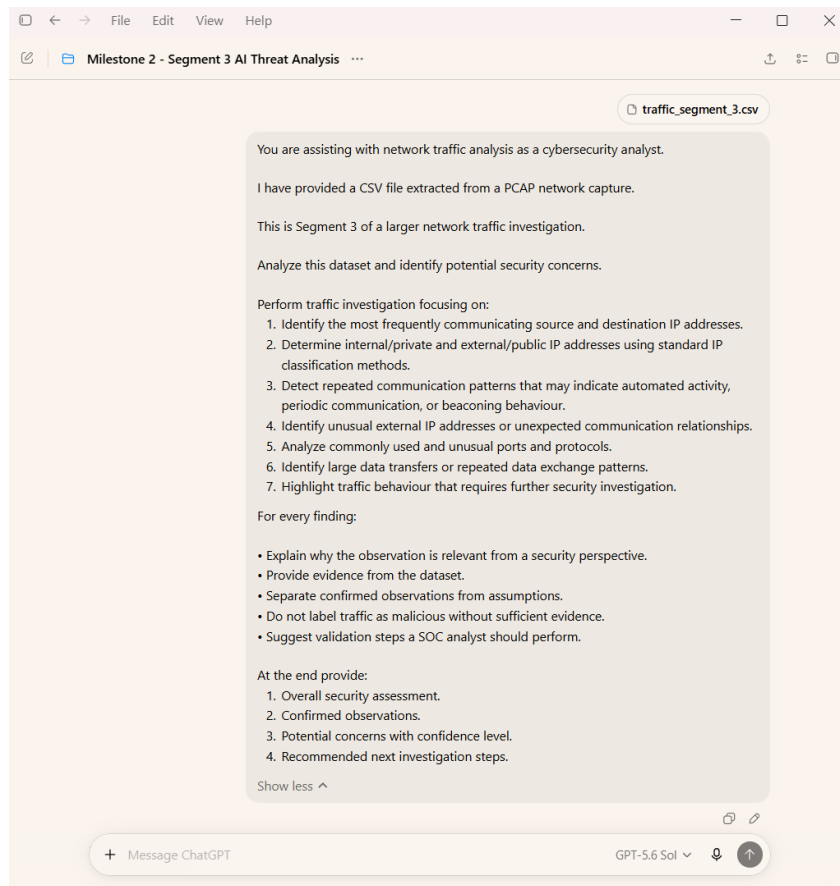


Figure 7: Segment 3 AI Investigation Prompt

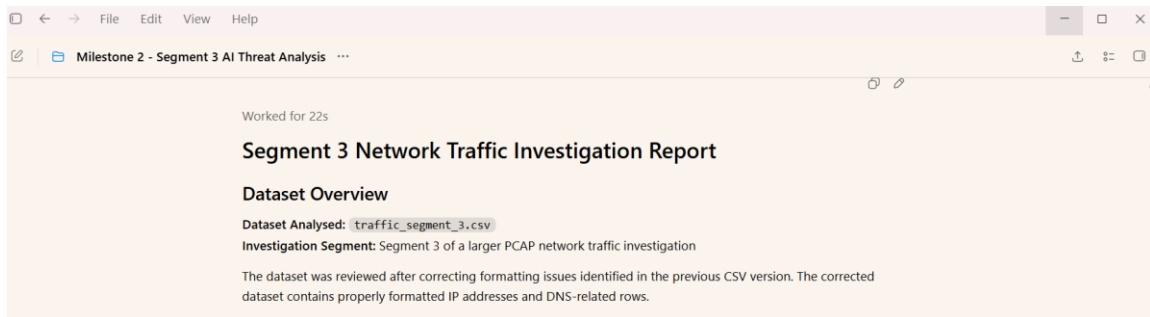


Figure 8: Segment 3 AI Generated Response

3. AI Analysis Evidence and Findings

3.1 Internal Host Identification

The AI consistently identified 10.10.0.33 as the primary internal communicating host across all three traffic segments.

Segment	Source Packets (10.10.0.33)	Destination Packets (10.10.0.33)
Segment 1	583	612
Segment 2	610	590
Segment 3	616	538

Classification: 10.10.0.33 falls within the RFC1918 private address range 10.0.0.0/8, confirming it as an internal host. Private IP addresses are used inside internal networks and are not directly routable on the public internet.

Security relevance: A highly active internal host is significant for investigation. Normal explanations include user workstation activity, web browsing, software updates, and cloud application use, while potential concerns needing validation include automated background processes, unauthorized applications, and malware communication.

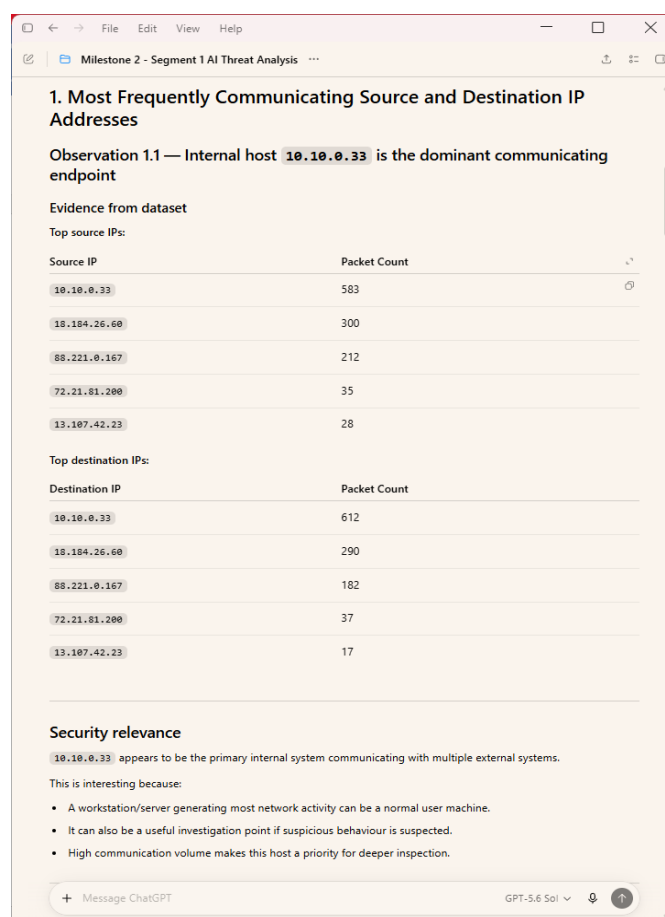


Figure 9: AI Key Finding, Internal Host

Evidence of the AI identifying the primary internal host and its communication volume.

3.2 External Communication Analysis

The AI identified several public external IP addresses communicating with 10.10.0.33. The main relationships observed were:

External IP Address	Segment Observed	Security Observation
18.184.26.60	Segment 1, Segment 2	Highest external communication volume
88.221.0.167	Segment 1	Repeated external communication
204.79.197.200	Segment 2, Segment 3	HTTPS/web communication
204.11.56.48	Segment 3	Repeated external communication
34.102.136.180	Segment 3	Cloud/web communication

External IP Address	Segment Observed	Security Observation
8.8.8.8	Segment 1, Segment 2, Segment 3	DNS communication

18.184.26.60: Dominant External Relationship

The strongest external relationship identified was between 10.10.0.33 and 18.184.26.60, observed in Segment 2: 575 packets were sent from 10.10.0.33 to 18.184.26.60 and 543 packets were returned, for a total of 1,118 packets, primarily over TCP port 80 (HTTP). Possible legitimate explanations include website communication, application services, software updates, and background application traffic; possible security concerns include unauthorized automated communication, malware-controlled communication, and suspicious application behaviour. Given the high packet volume, the dominance of a single external destination, and the use of plain HTTP, this pattern is assessed as medium investigation priority, though no malicious payload evidence or confirmed command-and-control behaviour was identified.

4. Unusual External IP Addresses and Communication Relationships

External Communication Summary

External IP	Packets	Main Port	Observation
18.184.26.60	1118	TCP/80	Dominant external communication
204.79.197.200	73	TCP/443	HTTPS communication
93.184.220.29	5	TCP/80	Small web communication
8.8.8.8	2	UDP/53	DNS query
13.107.42.23	1	Unknown TCP	Minimal communication

Finding 2: External IP 18.184.26.60

Evidence

Traffic:

Code: 10.10.0.33 → 18.184.26.60

Protocol:

Code: TCP

Destination port:

Code: 80

Packet count:

Code: 575 outbound, 543 inbound

Security Relevance

A single external server handling most communication from an internal system deserves investigation because:

- It represents a major external dependency.
- If unauthorized, compromise impact could be significant.

Figure 10: AI Key Finding, External Communication

Evidence of the dominant external communication relationship identified during analysis.

3.3 Port and Protocol Analysis

The AI analyzed the protocols and services observed across the three traffic segments. The main protocols identified were:

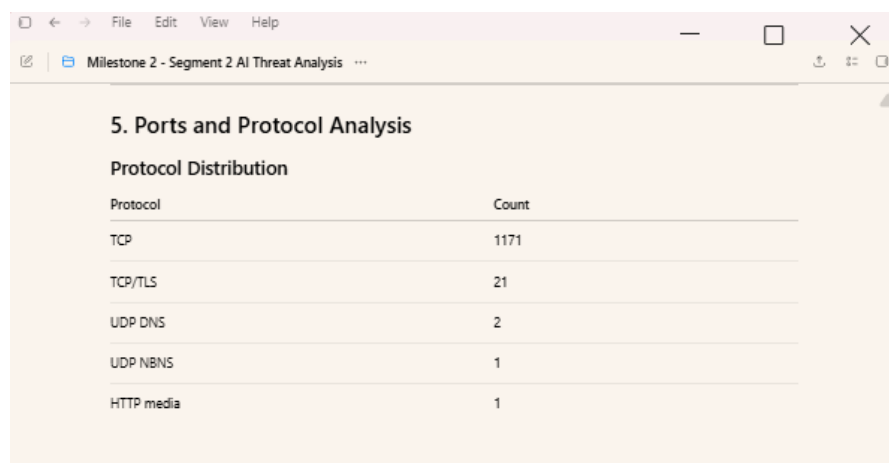
Protocol	Security Interpretation
TCP	Primary communication protocol
HTTP	Web communication over port 80
HTTPS/TLS	Encrypted web communication
DNS	Domain resolution traffic
NBNS	Windows network discovery traffic

TCP Port 80 (HTTP) and Port 443 (HTTPS/TLS)

In Segment 2, port 80 accounted for 577 packets, with the main communication flow running from 10.10.0.33:53480 to 18.184.26.60:80. HTTP traffic is not encrypted, so sensitive information could potentially be exposed and application communication may be visible to anyone observing the network path, though HTTP usage alone does not indicate malicious activity. Encrypted communication on port 443 was observed toward 204.79.197.200, among other destinations; HTTPS is commonly used for secure websites, cloud services, and application communication, and no suspicious HTTPS behaviour was confirmed.

UDP Port 53 (DNS) and Port 137 (NBNS)

DNS traffic on port 53 was observed toward 8.8.8.8, consistent with normal domain resolution activity. NBNS traffic on port 137 was observed directed at the internal broadcast address 10.10.0.255, consistent with routine Windows network discovery. This carries low security concern since it remained within the internal network and no exploitation behaviour was observed.



The screenshot shows a report window titled 'Milestone 2 - Segment 2 AI Threat Analysis'. The report content is titled '5. Ports and Protocol Analysis' and includes a section 'Protocol Distribution' with the following table:

Protocol	Count
TCP	1171
TCP/TLS	21
UDP DNS	2
UDP NBNS	1
HTTP media	1

Figure 11: AI Key Finding, Protocol Distribution

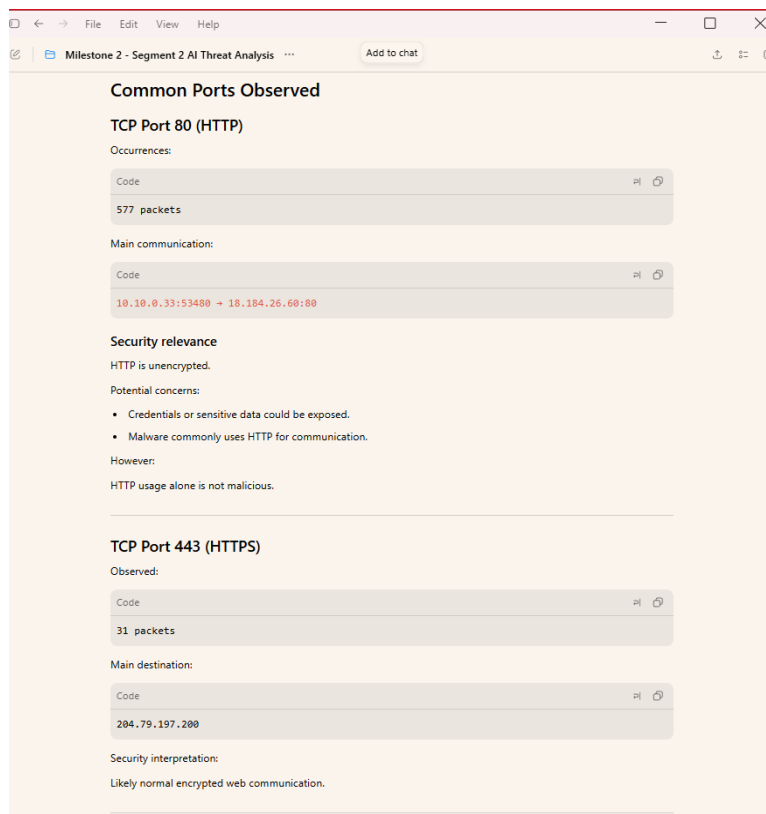


Figure 12: AI Key Finding, TCP Port Analysis

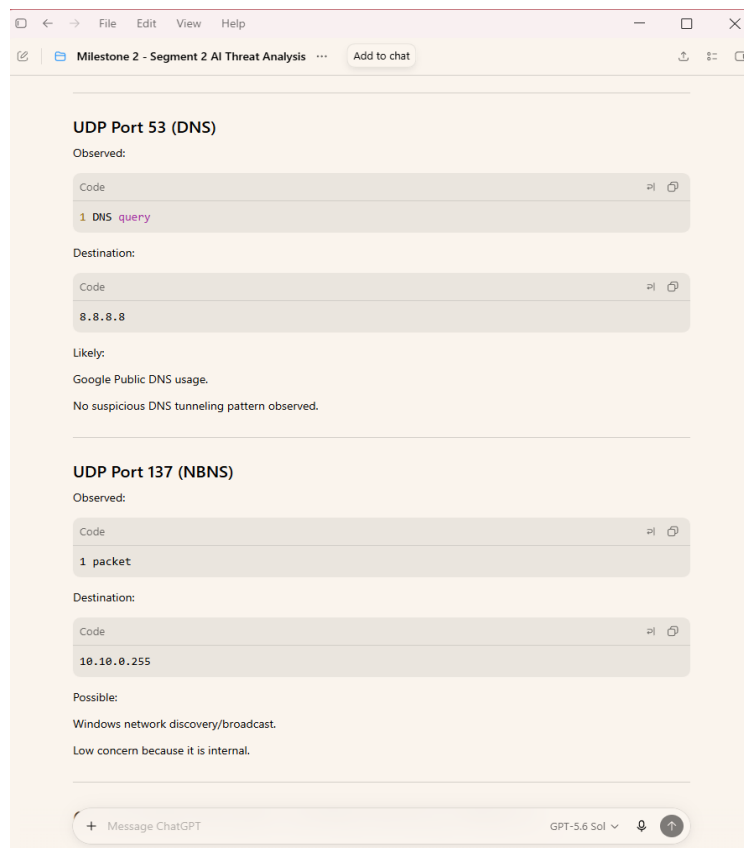


Figure 13: AI Key Finding, UDP Port Analysis

Evidence of the protocol distribution and the TCP and UDP port analysis discussed above.

3.4 DNS Large Frame and Data Exchange Analysis

Segment 3 analysis identified unusually large frame sizes in traffic from 10.10.0.33 to 8.8.8.8.

Source IP	Destination IP	Frame Length
10.10.0.33	8.8.8.8	58,337 bytes
10.10.0.33	8.8.8.8	54,032 bytes
10.10.0.33	8.8.8.8	53,529 bytes
10.10.0.33	8.8.8.8	52,160 bytes

Large DNS-related traffic volumes require validation, since unusual DNS activity can sometimes indicate abnormal application behaviour, large data exchanges, or potential DNS misuse. The observed frame sizes alone do not confirm DNS tunnelling, data exfiltration, or malware activity; additional packet-level analysis would be required. A SOC analyst should inspect the DNS query names, query length, DNS record types (A, AAAA, TXT, CNAME), randomized domain patterns, and query frequency.

6. Large Frame Size and Data Exchange Analysis

Largest Frame Sizes Observed

Source IP	Destination IP	Frame Length
10.10.0.33	8.8.8.8	58,337 bytes
10.10.0.33	8.8.8.8	54,032 bytes
10.10.0.33	8.8.8.8	53,529 bytes
10.10.0.33	8.8.8.8	52,160 bytes

Security Interpretation

Confirmed Observation

Large frame sizes were observed involving:

Code
10.10.0.33 → 8.8.8.8

The largest observed frame size was:

Code
58,337 bytes

Security Relevance

Large frame sizes may require additional validation because unusual traffic volumes can sometimes indicate:

- Application behaviour
- File transfer activity
- Abnormal communication patterns

However, frame size alone does not confirm:

- DNS tunnelling
- Data exfiltration
- Malicious activity

+ Message ChatGPT GPT-5.6 Sol

Figure 14: AI Key Finding, DNS Large Frame Analysis

Evidence of the large frame size observation identified in Segment 3.

4. Combined Security Assessment

After analysing all three segments, the same internal host, 10.10.0.33, appeared consistently as the main communication endpoint. It communicated with multiple external systems, including 18.184.26.60, 88.221.0.167, 204.79.197.200, 204.11.56.48, 34.102.136.180, and 8.8.8.8, all of which require review. Repeated communication patterns were also identified, including high packet exchange with 18.184.26.60 and multiple external service connections, though the traffic did not strongly match classic malware beaconing: no regular periodic callbacks were confirmed, and no malicious payload evidence was available.

Overall Risk Evaluation

Finding	Risk Level
Primary internal host communication	Medium
Dominant external communication	Medium
HTTP communication	Medium
Automated behaviour possibility	Low-Medium
DNS anomaly possibility	Low-Medium
Confirmed malware activity	Not confirmed

5. Confirmed Observations

The following findings were confirmed directly from the available CSV metadata:

- Primary internal host: 10.10.0.33, based on the highest communication frequency across all segments
- Major external communication: 18.184.26.60, with 1,118 packets exchanged in Segment 2
- Common network services identified: TCP, HTTP, HTTPS/TLS, DNS, NBNS
- Large frame sizes observed from 10.10.0.33 to 8.8.8.8, with a maximum of 58,337 bytes
- No evidence found confirming malware infection, command-and-control activity, data exfiltration, or unauthorized access

6. Attack Timeline Summary

An investigation timeline was created to represent the sequence of activities carried out during the network traffic analysis. It combines observations from CSV dataset preparation, AI-assisted traffic analysis, segment-level findings, and cross-segment correlation, and highlights communication events that require further security review.

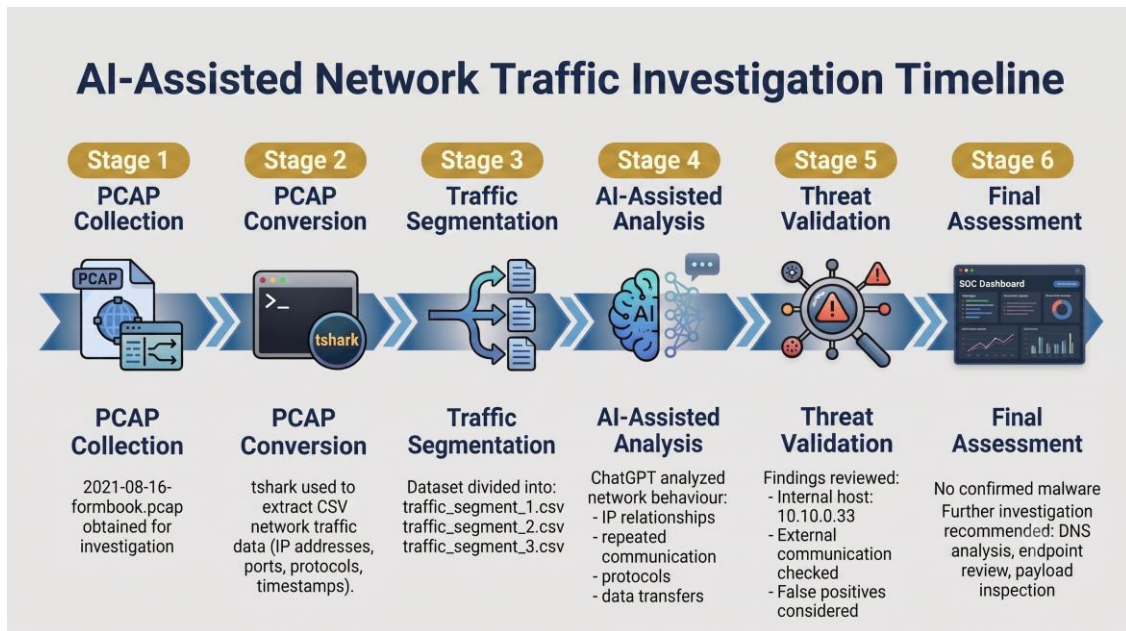


Figure 15: AI-Assisted Network Traffic Investigation Timeline

Visual summary of the investigation timeline listed below.

Stage	Activity	Observation
1	PCAP conversion	Original PCAP capture converted into CSV format for analysis
2	Dataset validation	CSV structure and packet records verified
3	Dataset segmentation	Traffic divided into three analysis segments
4	Segment 1 AI analysis	Identified internal host 10.10.0.33 and major external communication
5	Segment 2 AI analysis	Identified dominant HTTP communication with 18.184.26.60
6	Segment 3 AI analysis	Identified multiple external services and large DNS-related frame sizes
7	Cross-segment correlation	Compared common hosts, protocols, and communication patterns
8	Security assessment	Classified observations as requiring investigation rather than confirmed threats

7. SOC Analyst Investigation Steps

The AI-assisted analysis identified several areas requiring further validation. A SOC analyst should carry out the following investigation activities.

7.1 Investigate Internal Host Activity and External IP Reputation

For 10.10.0.33, recommended checks include identifying the hostname and asset owner, determining the operating system, reviewing installed applications, checking running processes, reviewing scheduled tasks, examining startup persistence mechanisms, and reviewing endpoint security alerts, since this host generated the highest volume of communication across the analyzed

traffic. Priority external IPs for reputation review are 18.184.26.60, 88.221.0.167, 204.79.197.200, 204.11.56.48, and 34.102.136.180, using WHOIS lookups, ASN ownership, threat intelligence databases, IP reputation services, and historical communication analysis to determine whether these destinations belong to legitimate services, cloud providers, or unknown infrastructure.

7.2 Perform Packet-Level and DNS Analysis

The CSV analysis was based primarily on metadata. Additional analysis should include reopening the original PCAP, following TCP streams, reviewing HTTP requests and responses, examining TLS certificates, and inspecting payload contents where available, since packet metadata alone cannot confirm malware communication, data theft, or command-and-control traffic. For the 10.10.0.33 to 8.8.8.8 relationship, review should also cover DNS query names, query frequency, domain reputation, TXT record usage, and long or randomized domain names, since DNS can sometimes be abused for covert communication or tunnelling, though none was confirmed from the available dataset.

7.3 Compare Against Normal Baseline

Recommended comparison points include previous network behaviour, firewall logs, proxy logs, endpoint telemetry, and user activity records, to determine whether the observed traffic is normal for the environment or unexpected for this specific host.

8. Mitigation Recommendations

Based on the AI-assisted investigation findings, the following security improvements are recommended.

8.1 Monitor and Restrict High-Value Host Traffic

Continuous monitoring should be implemented for important endpoints such as 10.10.0.33, including endpoint monitoring, review of unusual outbound connections, monitoring of new processes, and detection of abnormal network behaviour, to help identify compromised hosts or unauthorized applications earlier. Network access controls should also be applied to reduce unnecessary outbound traffic, using firewall outbound rules, allowing only required destinations, and monitoring unknown external connections, reducing the ability of malware to communicate externally.

8.2 Enforce Encryption and DNS Controls

HTTP communication was observed on TCP port 80. Where possible, HTTP should be replaced with HTTPS, encrypted communication should be enforced, and sensitive information should not be transmitted over unencrypted protocols, to protect data confidentiality. DNS monitoring should also be improved by logging DNS requests, monitoring suspicious domains, detecting unusual DNS volume, and blocking known malicious domains, helping detect DNS abuse, malware infrastructure, and suspicious domain activity.

8.3 Maintain Endpoint Security and Network Segmentation

Endpoints should maintain updated antivirus protection, endpoint detection and response (EDR) tooling, security monitoring, and regular patching, providing visibility into the processes generating network activity. Critical systems should also be separated from general user networks through

VLAN segmentation, restricted unnecessary internal communication, and least-privilege network access, limiting the impact of a compromised device.

9. Final Analyst Assessment

The AI-assisted investigation identified several network behaviours requiring security review. The most significant observations were the high communication activity of internal host 10.10.0.33 across all segments, the large external communication relationship with 18.184.26.60, the network protocols identified (TCP, HTTP, HTTPS/TLS, DNS, and NBNS), and the large frame sizes observed between 10.10.0.33 and 8.8.8.8, with a maximum size of 58,337 bytes.

Threat Classification

Category	Assessment
Confirmed malware	Not identified
Confirmed C2 communication	Not identified
Confirmed data exfiltration	Not identified
Suspicious communication requiring review	Identified
Further investigation required	Yes

The analyzed traffic shows significant communication activity between the internal host and external services, including high-volume communication, repeated patterns, HTTP-based traffic, and large DNS-related frame sizes that warrant further investigation. The available CSV metadata alone cannot confirm malicious activity; PCAP inspection, endpoint analysis, threat intelligence checks, DNS investigation, and security monitoring logs are required before a final determination can be made.

10. Conclusion

This milestone demonstrated the use of AI as an assisted cybersecurity analysis tool for investigating PCAP-derived network traffic, helping identify key communication relationships, hosts, protocol usage, and traffic patterns requiring further investigation. The work followed a security analyst workflow: preparing and validating the data, segmenting it, performing AI-assisted analysis, reviewing findings manually, and developing investigation recommendations. The final assessment is that the traffic contains investigation-worthy indicators but no confirmed evidence of malicious activity based on the available metadata alone; final security decisions still require additional validation using traditional SOC methods.